UNIVERSIDAD DEL ISTMO DE GUATEMALA

FACULTAD DE INGENIERIA

ELECTRONICA 2 Y DISENO ELECTRICO 1

PCIe PI7C9X3G808GP



KEVIN SEBASTIAN FLORES LOPEZ

OCTUBRE DE 2026

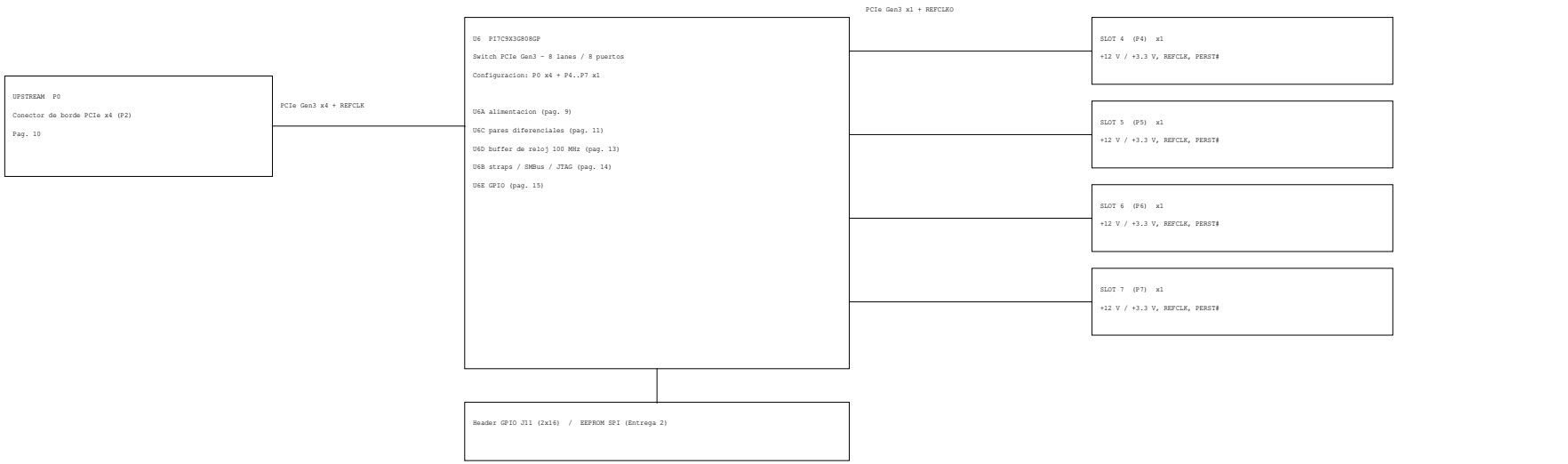
UNIS - Facultad de Ingeniería - Electronica 2		
Title: PORTADA		
Size: C	Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP	Rev: 1.0
Date: Thursday, October 29, 2026	Sheet 1 of 15	

INDICE DE HOJAS

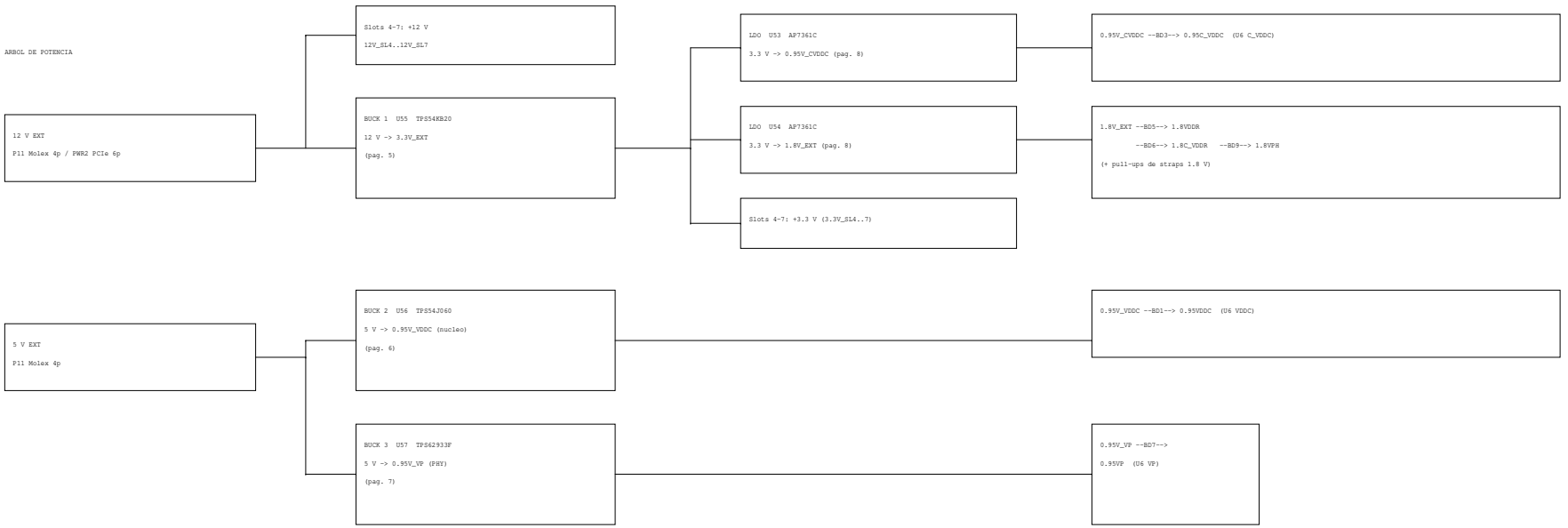
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DIAGRAMA DE BLOQUES Y ARBOL DE POTENCIA

DIAGRAMA DE BLOQUES (SENALES)



ARBOL DE POTENCIA

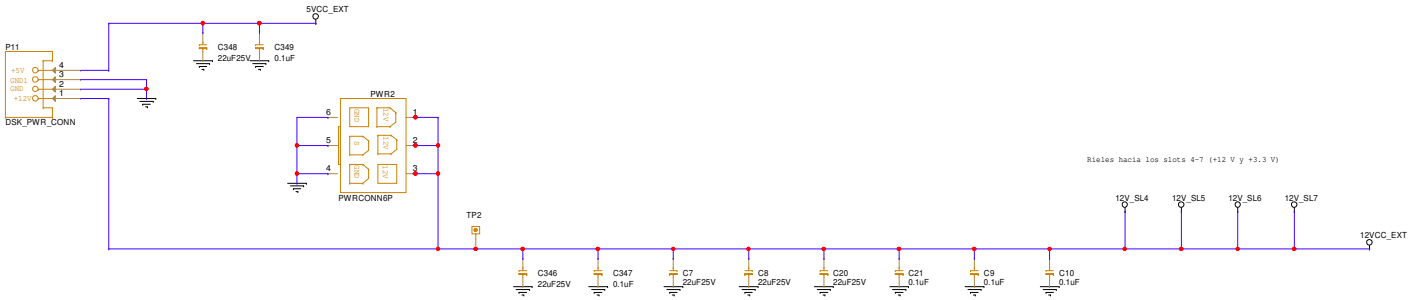


Secuencia de arranque: 3.3V_EXT (PG_3V3) -> EN Buck 2 -> PGOOD (U2_PGOOD) -> EN Buck 3. LEDs indicadores en pag. 4.

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Title: DIAGRAMA DE BLOQUES Y ARBOL DE POTENCIA		
Size: C	Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP	Rev: 1.0
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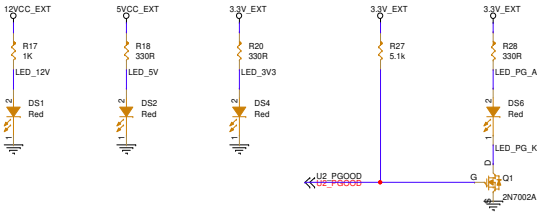
ENTRADA DE POTENCIA 12 V / 5 V

Entrada de 12 V y 5 V externos (P11 Molex 4p / PWR2 PCIe 6p), desacople bulk, distribucion a los slots 4-7 e indicadores LED.





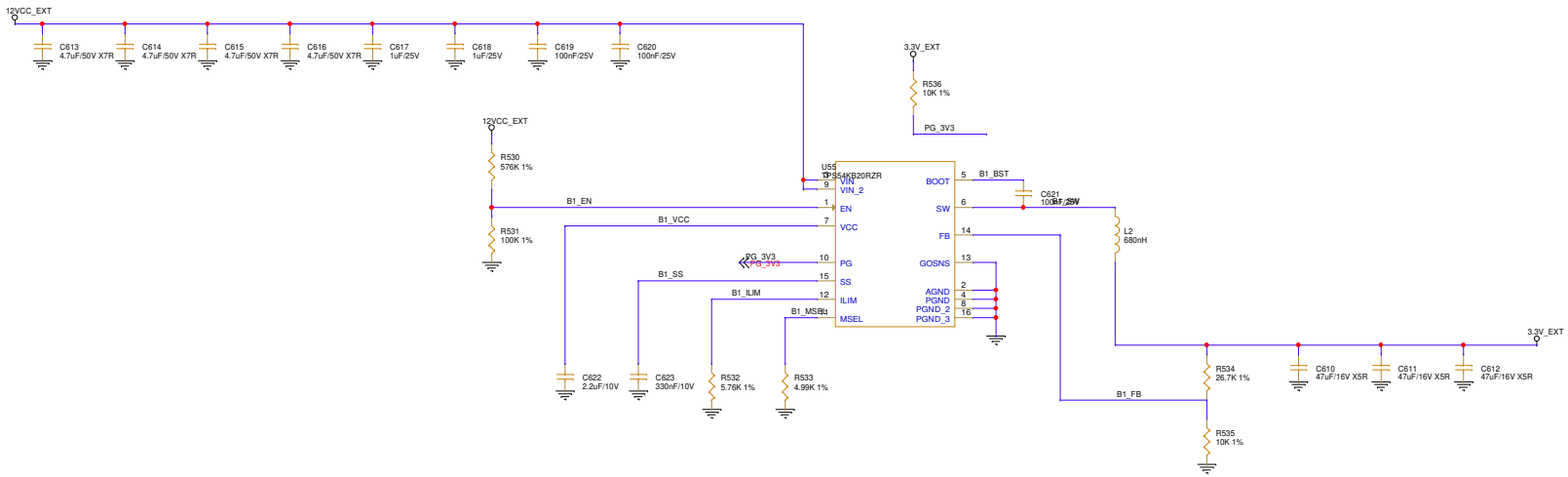
Indicadores: LED por riel y LED de PGOOD de 0.95V_VDDC (Q1)



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Title: ENTRADA DE POTENCIA 12 V / 5 V		
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BUCK 1: 12 V -> 3.3 V

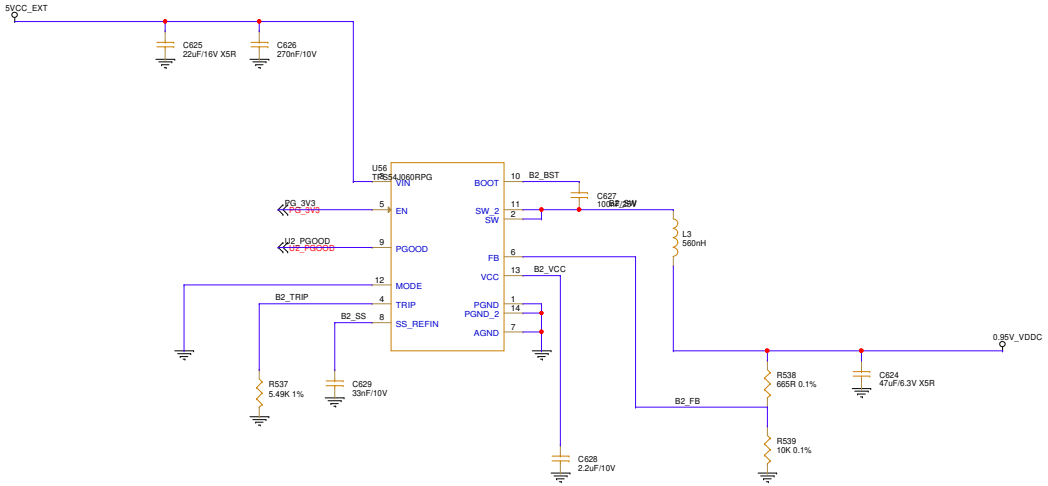
U55 TPS54KB20: 12 V -> 3.3V_EXT. Alimenta los slots (+3.3 V) y los LDO U53/U54. Su PG (PG_3V3) habilita el Buck 2.



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Title: BUCK 1: 12 V -> 3.3 V (TPS54KB20)		
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BUCK 2: 5 V -> 0.95 V VDDC

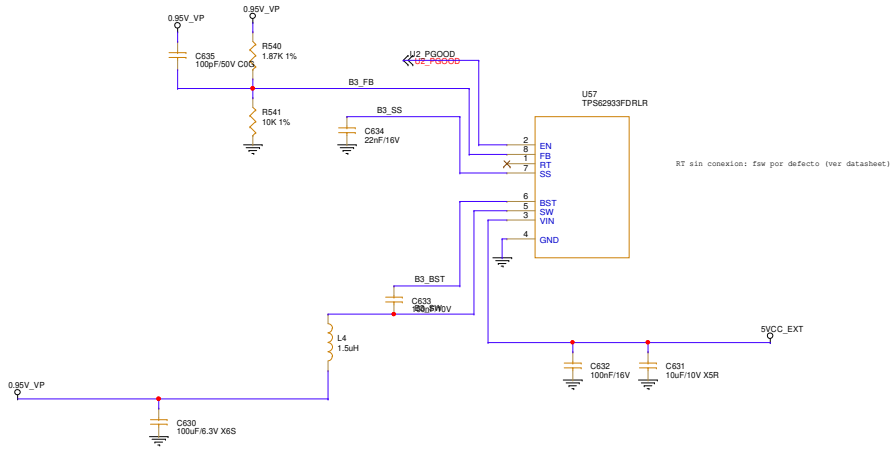
U56 TPS54J060: 5 V -> 0.95V_VDDC (nucleo del switch). EN = PG_3V3. Su PGOOD (U2_PGOOD) habilita el Buck 3 y enciende DS6.



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Title: BUCK 2: 5 V -> 0.95 V VDDC (TPS54J060)		
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BUCK 3: 5 V -> 0.95 V VP

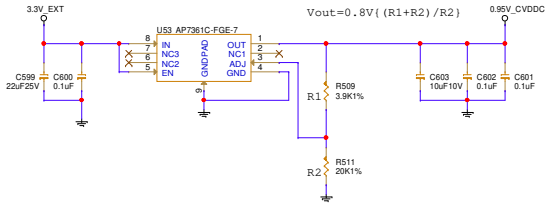
U57 TPS62933F: 5 V -> 0.95V_VP (PHY del switch). EN = U2_PGOOD. Vout = 0.8 V x (1 + R540/R541) = 0.95 V.



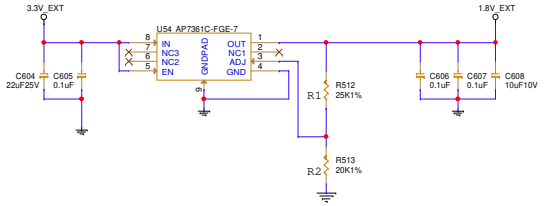
UNIS - Facultad de Ingeniería - Electronica 2		
Title: BUCK 3: 5 V -> 0.95 V VP (TPS62933F)		
Size: C	Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP	Rev: 1.0
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LDO 0.95 V C_VDDC / 1.8 V

0.95V / 1A

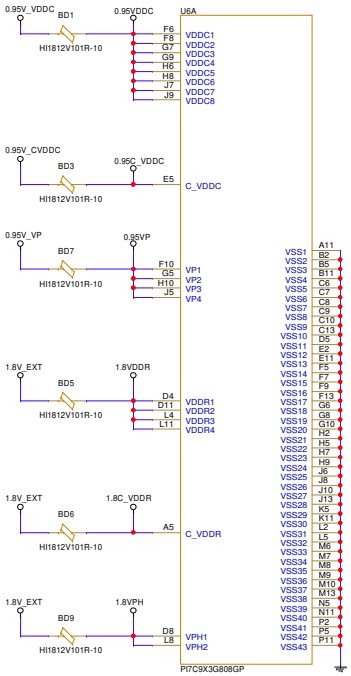


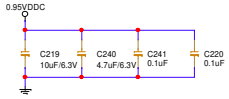
1.8V / 1A

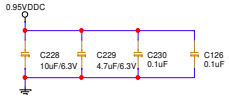


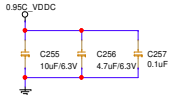
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Title: LDO 0.95 V C_VDDC Y 1.8 V (AP7361C)		
Size: C	Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP	Rev: 1.0
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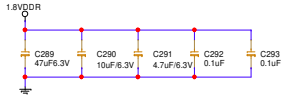
U6A – ALIMENTACION Y DESACOPLE

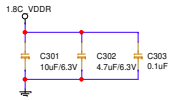


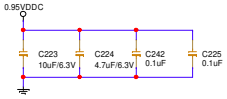


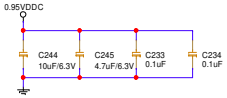


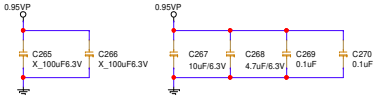


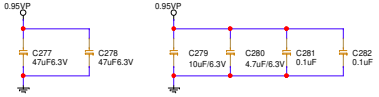


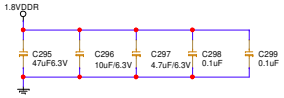


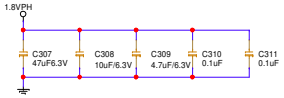












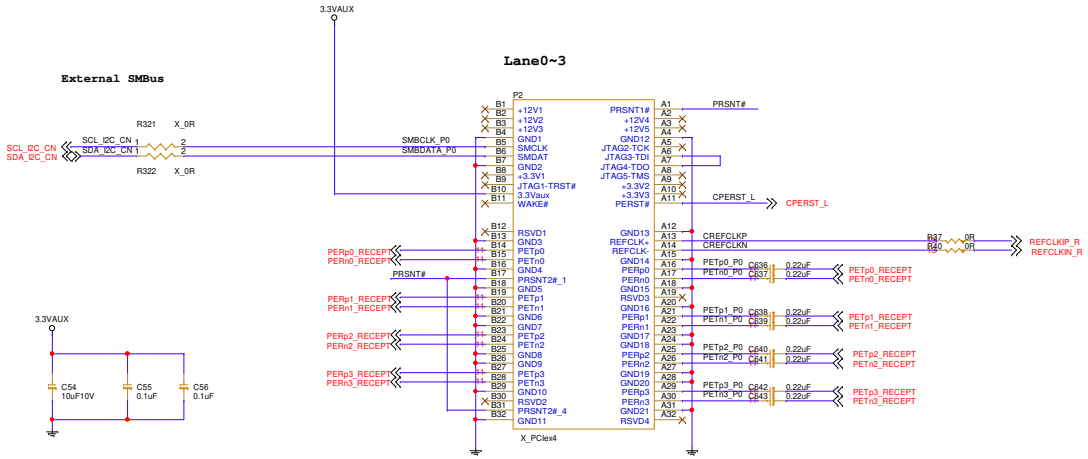
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Title: U6A: ALIMENTACION Y DESACOPLE DEL SWITCH

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C	PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP	1.0

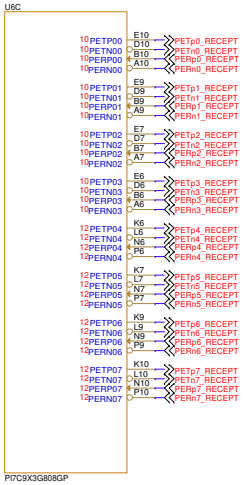
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UPSTREAM PCIe x4 (PUERTO 0)



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Title: UPSTREAM PCIe x4 (PUERTO 0)		
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U6C - PARES DIFERENCIALES



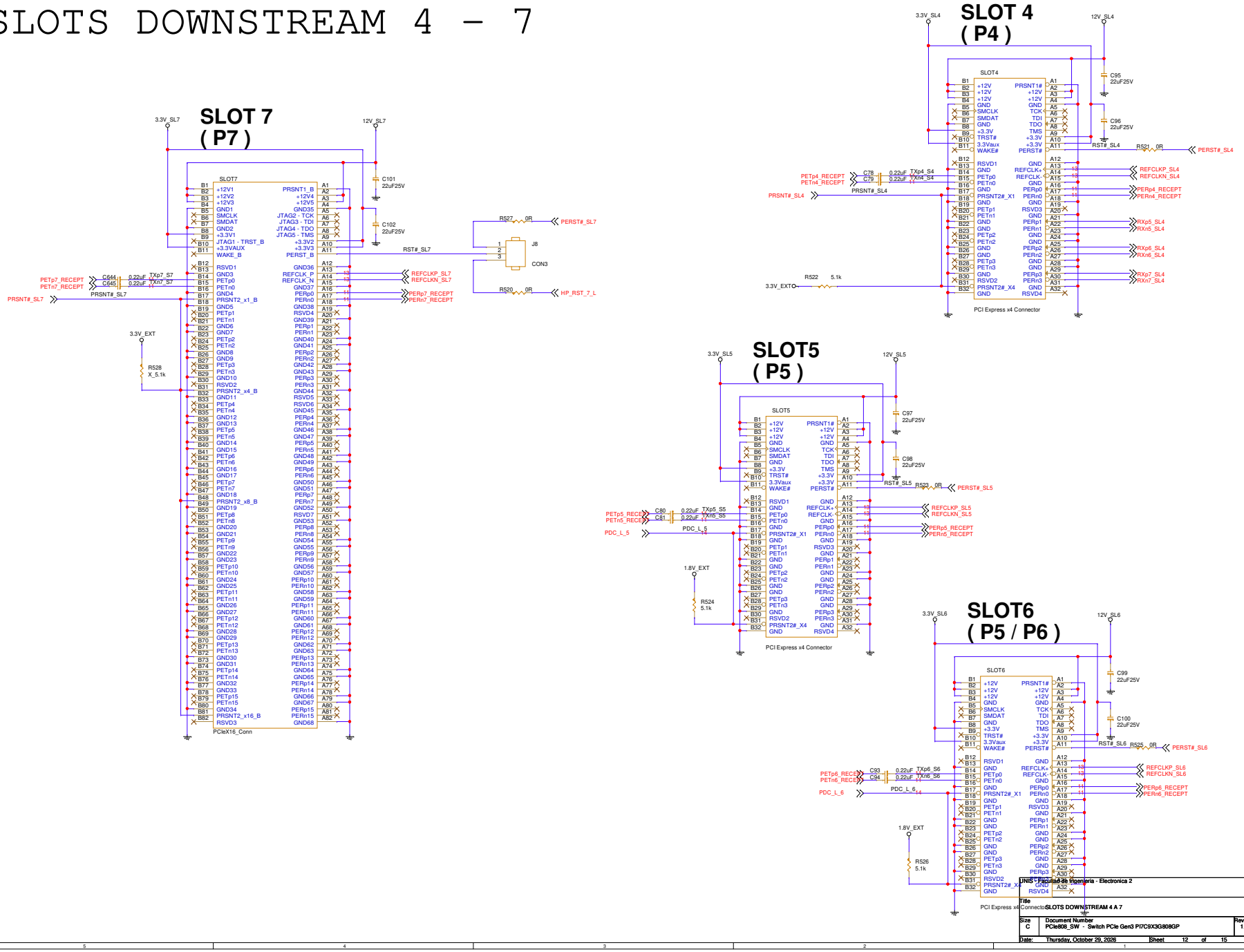
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Title: U6C: PARES DIFERENCIALES PCIe

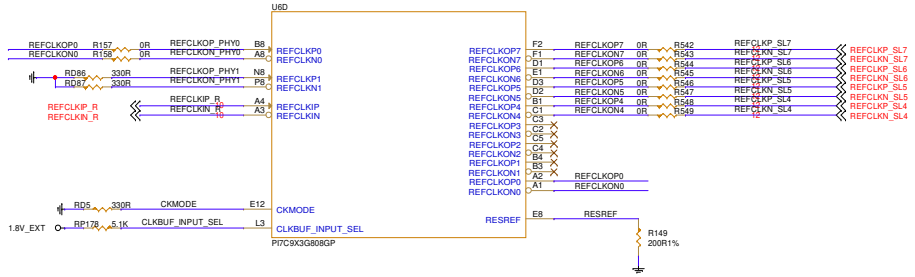
Size	Document Number	Rev
C	PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP	1.0

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SLOTS DOWNSTREAM 4 - 7



U6D – RELOJ DE REFERENCIA



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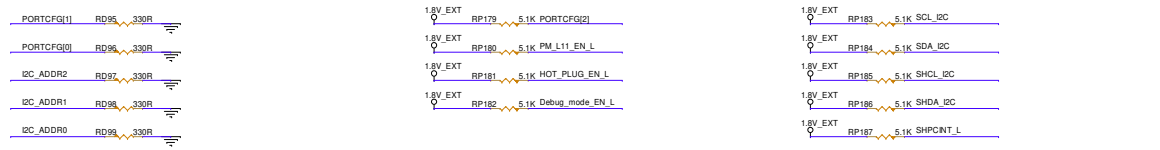
Title: U6D: RELOJ DE REFERENCIA

Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0

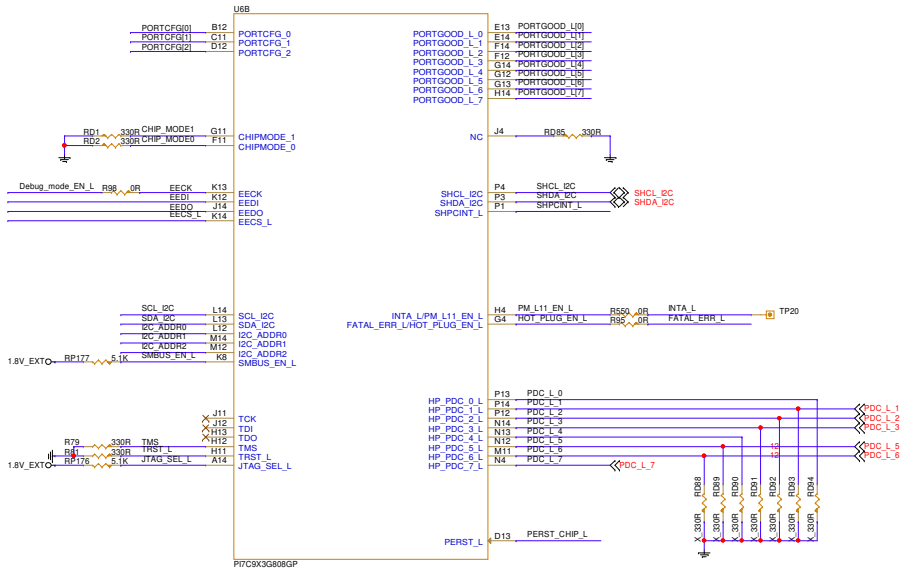
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U6B – CONFIGURACION Y CONTROL

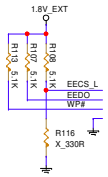
STRAPS DE CONFIGURACION (obligatorios, ver DS43471 sec. 3.2/3.5)



PORTCFG[2:0]=100b: modo 5 puertos (P0 x4 upstream, P4-P7 x1) | I2C_ADDR[2:0]=000 | PM L1.1 y Hot-Plug deshabilitados | Debug mode deshabilitado



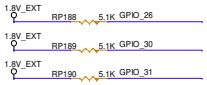
PULL-UPS DE LA INTERFAZ EEPROM SPI (la EEPROM se agrega en la Entrega 2). R116 = DNP

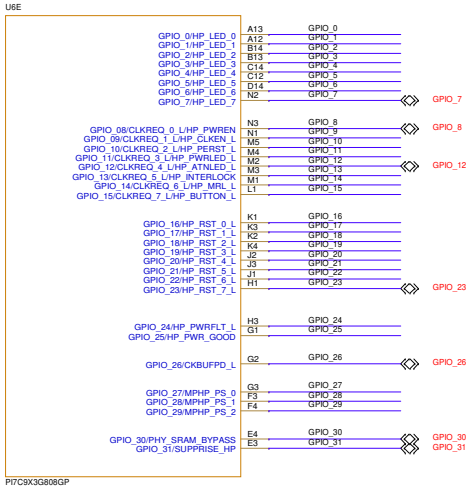


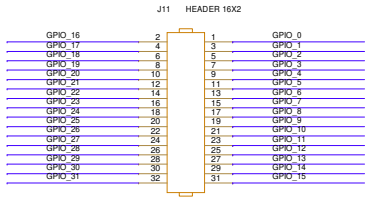
UNIS - Facultad de Ingeniería - Electronica 2		
Title: U6B: CONFIGURACION, STRAPS, SMBus Y JTAG		
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U6E - GPIO

PULL-UPS DE GPIO CON FUNCION ALTERNA: GPIO26=CKBUFPD_L (buffer de reloj activo), GPIO30=PHY_SRAM_BYPASS, GPIO31=SURPRISE_HP







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Title: U6E: GPIO Y HEADER J11		
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